Synthetic Equivalents Exercise

The six basic synthetic equivalents:

long call + short put ≈ ***synthetic long underlying***

short call + long put ≈ ***synthetic short underlying***

long call + short underlying ≈ ***synthetic long put***

short call + short underlying ≈ ***synthetic short put***

long put + long underlying ≈ ***synthetic long call***

short put + short underlying ≈ ***synthetic short call***

All options must have the same exercise price and expiration date

Synthetic Equivalents Exercise

For each combination position below, what is the synthetic equivalent or strategy.

a.) -1 January 70 put / -1 underlying contract

b.) -1 February 65 call / -1 February 65 put

c.) -1 March 75 call / +1 underlying contract

d.) -1 April 225 put / +1 April 175 put

e.) +1 underlying contract / +1 May 250 put

f.) +1 June 200 call / +1 June 200 put

g.) +1 July 130 call / -1 July 130 put

h.) +1 August 115 call / -1 underlying contract

i.) -1 September 100 put / -1 September 110 call

j.) -1 October 115 call / +1 October 115 put

Synthetic Pricing Exercise

Basic Put-Call-Parity: C - P = S - X

1. Assume that interest rates and dividends are zero

a.) Fill in the missing call and put prices

stock price = 61.75

exercise price	45	50	55	60	65	70	75	80
call price		11.90		4.05		.70		.10
put price	.05		.75		5.10		13.50	

b.) For each option below, fill in the missing value

stock price	exercise price	call price	put price
152.65	140	15.80	
	475	6.99	20.48
80.87		4.48	3.61
26.77	25		.22

Synthetic Pricing Exercise

2. Options on stock

put-call parity for stock options $C - P = (F - X) / (1 + r \times t)$

where $F = S \times (1 + r \times t) - D$

Rather than use the full form, stock option traders sometimes use this approximation in order to simplify the arithmetic: $C - P \approx S - X + X \times r \times t - D$

In the following questions you may use either the full form or the approximation. The approximation may result in answers slightly different than those given.

a.) stock price = 150.10

time to expiration = 2 months

interest rate = 3.00%

expected dividends = .85

exercise price	130	135	140	145	150	155	160	165
call price		16.30		9.25		4.45		1.85
put price	.70		2.55		6.55		12.90	

Synthetic Pricing Exercise

2. Options on stock

put-call parity for stock options $C - P = (F - X) / (1 + r \times t)$

or $C - P \approx S - X + X \times r \times t - D$

b.) For each option below, fill in the missing value

stock price	exercise price	time to expiration (days)	interest rate	expected dividends	call price	put price
260.03	270	34	5.69%	2.00		14.04
23.97	25	137	3.32%		1.31	2.20
446.75	450	96		1.75	17.30	17.30
86.05	80	73	4.50%	.62	8.27	
	125	64	8.00%	.95	5.72	3.93
59.15	50		2.71%	0	10.70	.70

Synthetic Equivalents Exercise (answers)

For each combination position below, what is the synthetic equivalent or strategy.

a.)	-1 January 70 put / -1 underlying contract	***-1 January 70 call***
b.)	-1 February 65 call / -1 February 65 put	***short straddle***
c.)	-1 March 75 call / +1 underlying contract	***-1 March 75 put***
d.)	-1 April 225 put / +1 April 175 put	***bear spread***
e.)	+1 underlying contract / +1 May 250 put	***-1 May 250 call***
f.)	+1 June 200 call / +1 June 200 put	***long straddle***
g.)	+1 July 130 call / -1 July 130 put	***+1 underlying contract***
h.)	+1 August 115 call / -1 underlying contract	***+1 August 115 put***
i.)	-1 September 100 put / -1 September 110 call	***short strangle***
j.)	-1 October 115 call / +1 October 115 put	***-1 underlying contract***

Synthetic Pricing Exercise (answers)

Basic Put-Call-Parity: C - P = S - X

1. Assume that interest rates and dividends are zero

   a.) Fill in the missing call and put prices

   stock price = 61.75

exercise price	45	50	55	60	65	70	75	80
call price	***16.80***	11.90	***7.50***	4.05	***1.85***	.70	***.25***	.10
put price	.05	***.15***	.75	***2.30***	5.10	***8.95***	13.50	***18.35***

   b.) For each option below, fill in the missing value

stock price	exercise price	call price	put price
152.65	140	15.80	***3.15***
461.51	475	6.99	20.48
80.87	***80***	4.48	3.61
26.77	25	***1.99***	.22

Synthetic Pricing Exercise (answers)

2. Options on stock

put-call parity for stock options $C - P = (F - X) / (1 + r \times t)$

where $F = S \times (1 + r \times t) - D$

Rather than use the full form, stock option traders sometimes use this approximation in order to simplify the arithmetic: $C - P \approx S - X + X \times r \times t - D$

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a.) stock price = 150.10
time to expiration = 2 months
interest rate = 3.00%
expected dividends = .85

exercise price	130	135	140	145	150	155	160	165
call price	***20.60***	16.30	***12.50***	9.25	***6.55***	4.45	***2.95***	1.85
put price	.70	***1.37***	2.55	***4.27***	6.55	***9.43***	12.90	***16.78***

Synthetic Pricing Exercise (answers)

2. Options on stock

put-call parity for stock options $C - P = (F - X) / (1 + r \times t)$

or $C - P \approx S - X + X \times r \times t - D$

b.) For each option below, fill in the missing value

stock price	exercise price	time to expiration (days)	interest rate	expected dividends	call price	put price
260.03	270	34	5.69%	2.00	***3.50***	14.04
23.97	25	137	3.32%	***.17***	1.31	2.20
446.75	450	96	***4.26%***	1.75	17.30	17.30
86.05	80	73	4.50%	.62	8.27	***2.12***
126.00	125	64	8.00%	.95	5.72	3.93
59.15	50	***2.33***	2.71%	0	10.70	.70